

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                                               |
|-----------------------------|---------------------------------------------------------------|
| <b>Product Name</b>         | <u>Chloroform, HPLC Grade</u>                                 |
| <b>CAS No</b>               | 67-66-3                                                       |
| <b>Synonyms</b>             | Methane trichloride; Methenyl trichloride; Formyl trichloride |
| <b>Molecular Formula</b>    | C H Cl <sub>3</sub>                                           |
| <b>Molecular Weight</b>     | 119.38                                                        |
| <b>Recommended Use</b>      | Laboratory chemicals.                                         |
| <b>Uses advised against</b> | All other uses                                                |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>43685</b>                                                                                |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <u>ANZinfo@thermofisher.com</u>                                                             |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

**HSNO Approval Number**      **HSR002937**

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                                      |            |
|------------------------------------------------------|------------|
| Acute Oral Toxicity                                  | Category 4 |
| Acute Inhalation Toxicity - Vapors                   | Category 3 |
| Skin Corrosion/Irritation                            | Category 2 |
| Serious Eye Damage/Eye Irritation                    | Category 2 |
| Carcinogenicity                                      | Category 2 |
| Reproductive Toxicity                                | Category 2 |
| Specific target organ toxicity - (single exposure)   | Category 3 |
| Specific target organ toxicity - (repeated exposure) | Category 1 |

#### Environmental hazards

Chronic aquatic toxicity

Category 4

**Label Elements****Signal Word****Danger****Hazard Statements**

H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H331 - Toxic if inhaled  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer  
H361 - Suspected of damaging fertility or the unborn child  
H372 - Causes damage to organs through prolonged or repeated exposure  
H413 - May cause long lasting harmful effects to aquatic life

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection

**Response**

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P330 - Rinse mouth  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Cardiac and respiratory depression  
Overexposure may cause decreased heart rate, decreased blood pressure, heart block, and cardiac failure Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component  | CAS No  | Weight % |
|------------|---------|----------|
| Chloroform | 67-66-3 | >99      |

|           |          |      |
|-----------|----------|------|
| 1-Pentene | 109-67-1 | 0.01 |
|-----------|----------|------|

## Section 4 - First Aid Measures

### Description of first aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                     |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Self-Protection of the First Aider</b>  | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                 |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                           |
| <b>Most important symptoms and effects</b> | . Symptoms of overexposure are dizziness, headache, tiredness, nausea, unconsciousness, cessation of breathing: Causes central nervous system depression                                                                                                                                                                       |
| <b>Notes to Physician</b>                  | Treat symptomatically. Signs of overdose include stupor and respiratory depression. Symptoms may be delayed.                                                                                                                                                                                                                   |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Phosgene, Hydrogen chloride gas.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from direct sunlight. Store under an inert atmosphere. Protect from moisture.

**Incompatible Materials**

Strong oxidizing agents. Alkali metals. Aluminium. Acetone.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

| Component  | New Zealand WEL                                  | Australia                               | ACGIH TLV   | The United Kingdom                                                                      |
|------------|--------------------------------------------------|-----------------------------------------|-------------|-----------------------------------------------------------------------------------------|
| Chloroform | TWA: 2 ppm<br>TWA: 9.9 mg/m <sup>3</sup><br>Skin | TWA: 2 ppm<br>TWA: 10 mg/m <sup>3</sup> | TWA: 10 ppm | TWA: 2 ppm<br>TWA: 9.9 mg/m <sup>3</sup><br>STEL: 6 ppm<br>STEL: 29.7 mg/m <sup>3</sup> |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Appropriate engineering controls**Engineering Measures**

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Individual protection measures, such as personal protective equipment

**Eye Protection** Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                 |
|----------------|-------------------|-----------------|-----------------|--------------------------------------------------------------------------------|
| Viton (R).     | > 480 minutes     | -               | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Neoprene       | < 25 minutes      | 0.45 mm         |                 |                                                                                |
| Butyl rubber   | < 15 minutes      | 0.35 mm         |                 |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 (or AUS/NZ equivalent)  
**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
 When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains.

## Section 9 - Physical and Chemical Properties

Information on basic physical and chemical properties

|                                  |                          |                                          |
|----------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>            | Liquid                   |                                          |
| <b>Appearance</b>                | Colorless                |                                          |
| <b>Odor</b>                      | aromatic sweet           |                                          |
| <b>Odor Threshold</b>            | No data available        |                                          |
| <b>pH</b>                        | No information available |                                          |
| <b>Melting Point/Range</b>       | -63 °C / -81.4 °F        |                                          |
| <b>Softening Point</b>           | No data available        |                                          |
| <b>Boiling Point/Range</b>       | 61 °C / 141.8 °F         |                                          |
| <b>Flammability (liquid)</b>     | No data available        |                                          |
| <b>Flammability (solid,gas)</b>  | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>          | No data available        |                                          |
| <b>Flash Point</b>               | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>  | No data available        |                                          |
| <b>Decomposition Temperature</b> | No data available        |                                          |

|                                         |                          |             |
|-----------------------------------------|--------------------------|-------------|
| Viscosity                               | 0.56 mPa s at 20 °C      |             |
| Water Solubility                        | 8 g/L (20°C)             |             |
| Solubility in other solvents            | No information available |             |
| Partition Coefficient (n-octanol/water) |                          |             |
| Component                               | log Pow                  |             |
| Chloroform                              | 2                        |             |
| 1-Pentene                               | 2.66                     |             |
| Vapor Pressure                          | 213 mbar @ 20 °C         |             |
| Density / Specific Gravity              | 1.480                    |             |
| Bulk Density                            | Not applicable           | Liquid      |
| Vapor Density                           | No data available        | (Air = 1.0) |
| Particle characteristics                | Not applicable (liquid)  |             |

**Other information**

|                   |                     |
|-------------------|---------------------|
| Molecular Formula | C H Cl <sub>3</sub> |
| Molecular Weight  | 119.38              |
| VOC Content(%)    | 100                 |

## Section 10 - Stability and Reactivity

|                                  |                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                                             |
| Stability                        | Stable under normal conditions. UNSTABLE (REACTIVE) UPON DEPLETION OF INHIBITOR. Light sensitive.      |
| Sensitivity to Mechanical Impact | No information available                                                                               |
| Sensitivity to Static Discharge  | No information available                                                                               |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                               |
| Hazardous Reactions              | None under normal processing.                                                                          |
| Conditions to Avoid              | Incompatible products, Heat, flames and sparks, Excess heat, Exposure to light, Protect from moisture. |
| Incompatible Materials           | Strong oxidizing agents, Alkali metals, Aluminium, Acetone.                                            |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Phosgene. Hydrogen chloride gas.              |

## Section 11 - Toxicological Information

**Acute Effects****Information on likely routes of exposure****Product Information**

|            |                                                                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation | Irritating to respiratory system. May be harmful if inhaled. May cause irritation of respiratory tract.                                                   |
| Eyes       | Irritating to eyes. Contact with eyes may cause irritation.                                                                                               |
| Skin       | Irritating to skin. May be harmful in contact with skin. May cause eye/skin irritation.                                                                   |
| Ingestion  | Harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. Ingestion may cause irritation to mucous membranes. |

**Numerical measures of toxicity**

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| (a) acute toxicity; |                                                                  |
| Oral                | Category 4                                                       |
| Dermal              | Based on available data, the classification criteria are not met |

## Inhalation

Category 3

| Component  | LD50 Oral                                                                      | LD50 Dermal               | LC50 Inhalation              |
|------------|--------------------------------------------------------------------------------|---------------------------|------------------------------|
| Chloroform | LD50 = 908 mg/kg (rat)<br>LD50 = 695 mg/kg ( Rat )<br>LD50 = 450 mg/kg ( Rat ) | LD50 > 20 g/kg ( Rabbit ) | LC50 = 10.5 mg/L ( Rat ) 4 h |
| 1-Pentene  | >2000 mg/kg (Rat)                                                              | >2000 mg/kg (Rabbit)      | LC50 = 10000 ppm ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component  | New Zealand                | Australia            | New South Wales | Western Australia | IARC     | EU | UK | Germany |
|------------|----------------------------|----------------------|-----------------|-------------------|----------|----|----|---------|
| Chloroform | Suspected human carcinogen | suspected carcinogen |                 |                   | Group 2B |    |    |         |

(g) reproductive toxicity;  
Reproductive Effects  
Developmental Effects  
Teratogenicity

Category 2

Experiments have shown reproductive toxicity effects on laboratory animals

Developmental effects have occurred in experimental animals

Study result negative

(h) STOT-single exposure; Category 3

Results / Target organs

Central nervous system (CNS)

(i) STOT-repeated exposure; Category 1

Study result

LOAEL = 15 mg/kg bw/day

NOAEC = 25 mg/m<sup>3</sup>

Route of exposure

Inhalation

Target Organs

Liver, Kidney.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Other Adverse Effects

Tumorigenic effects have been reported in experimental animals.

Symptoms / effects, both acute and delayed

Symptoms of overexposure are dizziness, headache, tiredness, nausea, unconsciousness, cessation of breathing. Causes central nervous system depression.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity**

Do not empty into drains. The product contains following substances which are hazardous for the environment. Contains a substance which is: Harmful to aquatic organisms.

| Component  | Freshwater Fish                                                                                                                                                                                                                     | Water Flea           | Freshwater Algae    | Microtox                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chloroform | LC50: = 300 mg/L, 96h static (Poecilia reticulata)<br>LC50: = 18 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: = 18 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: = 71 mg/L, 96h flow-through (Pimephales promelas) | EC50 = 28.9 mg/L/48h | EC50 = 560 mg/L/48h | Photobacterium phosphoreum: EC50 = 520 mg/L/5 min<br>Photobacterium phosphoreum: EC50 = 670 mg/L/15 min<br>Photobacterium phosphoreum: EC50 = 670 mg/L/30min |

**Terrestrial ecotoxicity**

There is no data for this product

**Persistence and Degradability**

Product is biodegradable

**Persistence**

Persistence is unlikely, based on information available.

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component  | log Pow | Bioconcentration factor (BCF) |
|------------|---------|-------------------------------|
| Chloroform | 2       | 1.4 - 13 dimensionless        |
| 1-Pentene  | 2.66    | No data available             |

**Mobility**

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air

**Other adverse effects**
**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods****Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## Section 14 - Transport Information

| Component                      | Hazchem Code |
|--------------------------------|--------------|
| Chloroform<br>67-66-3 ( >99 )  | 2Z           |
| 1-Pentene<br>109-67-1 ( 0.01 ) | 3YE          |

### NZS 5433:2020

UN-No UN1888  
 Proper Shipping Name Chloroform  
 Hazard Class 6.1  
 Packing Group III

### IATA

UN-No UN1888  
 Proper Shipping Name Chloroform  
 Hazard Class 6.1  
 Packing Group III

### IMDG/IMO

UN-No UN1888  
 Proper Shipping Name Chloroform  
 Hazard Class 6.1  
 Packing Group III

Environmental hazards No hazards identified

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code Not applicable, packaged goods

Special Precautions No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                      |           |
|----------------------|-----------|
| HSNO Approval Number | HSR002937 |
|----------------------|-----------|

### **National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

|           |             |
|-----------|-------------|
| Component | New Zealand |
|-----------|-------------|

|            |                            |
|------------|----------------------------|
| Chloroform | Suspected human carcinogen |
|------------|----------------------------|

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions according to EU REACH**

| Component  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                           | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Chloroform | -                                                                   | Use restricted. See item 32.<br>(see <a href="http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32006R1907:EN:NOT">http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32006R1907:EN:NOT</a> for restriction details) | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component  | CAS No   | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|------------|----------|-------|------|-----------|--------|-----|----------|-------|------|
| Chloroform | 67-66-3  | X     | X    | 200-663-8 | -      | -   | X        | X     | X    |
| 1-Pentene  | 109-67-1 | X     | X    | 203-694-5 | -      | -   | KE-28027 | X     | X    |

| Component  | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|------------|----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Chloroform | 67-66-3  | X    | ACTIVE                                        | X   | -    | X     | X    | X    |
| 1-Pentene  | 109-67-1 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**Section 16 - Other Information**

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

**Legend**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

#### Revision Date

14-Mar-2023

#### Revision Summary

Not applicable

#### Disclaimer

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## End of Safety Data Sheet